Business Analytics: From Data to Decision

Organized by the CRISP-DM process

Phase 1. Business Understanding

- Types of analytics: descriptive, diagnostic, predictive, and prescriptive
- Translating a business question into an analytics problem
- Supervised vs. unsupervised learning (which one fits the question?)

Phase 2. Data Understanding

- Exploratory Data Analysis (EDA)
- Descriptive statistics
- Visualization to detect patterns, trends, and anomalies

Phase 3. Data Preparation

- Train/test split (done first, to avoid *data leakage*)
- Data cleaning and missing values
- Outliers
- Dummy variables
- Feature scaling
- Key rule: all transformations are calculated on the training set only, then applied to the test set

Phase 4. Modeling

- *Supervised models*
    - Regression (numeric outcome): simple linear, multiple linear
    - Classification (categorical outcome): logistic regression (it classifies despite its name), decision trees
- *Unsupervised models*
    - K-Means clustering
    - Choosing K: elbow method and silhouette
    - Why scaling matters for distance-based methods
- *Time series*
    - Data patterns: trend, seasonality, cycles, and randomness
    - Moving averages
    - Exponential smoothing: simple, Holt, and Holt-Winters
    - Time-based split: train on the past, test on the future (no random split)

Phase 5. Evaluation

- Predicted vs. actual values
- Forecast error
- Overfitting: why a "perfect" model can fail on new data
- Regression metrics: R^2, MAE, MSE/RMSE
- Classification metrics: confusion matrix, accuracy, precision, and recall
- Time series metrics: MAE, RMSE, and MAPE
- Business evaluation: does the model actually answer the original question?

Phase 6. Deployment

- Prescriptive analytics: optimization and simulation
- Communicating results effectively
- Monitoring the model over time